## LESSON 1: OVERVIEW OF HUMAN AND COMPUTER INTERACTION

### Human Computer Interaction

- is a field of study focused on **interfaces between users and computers**. HCI *observes the way how humans associate with computers to produce efficient and useful interaction, examining and observing some factors based on how humans use technology and establish designs and programs to enable applicable systems.*

### User Interaction Design

- Interaction design is about creating interventions in often complex situations using technology of many kinds including PC software, the web and physical devices.

**Design involves**:

- Achieving goals within constraints and trade-off
- Understanding the raw materials: computer and human
- Accepting limitations of human of design

**The design process has several stages and is iterative and never complete.**

**Interaction starts with getting to know the users and their context:**

- **Finding out who they are and what they are like …**
- **Talking to them, watching them**

- **Scenarios** are rich design stories, which can be used and reused throughout design:
  - *They help us see what users will want to do*
  - *They give a step-by-step walkthrough of users interactions: including what they see, do and are thinking.*
- Users need to find their way around a system; this involves:

  - helping users know where they are, where they have been and what they can do next.

  - creating overall structures that are easy to understand and fit the users needs.

  - designing comprehensible screens and control panels

- **Complexity of design** means we don't get it right first time:
  - So, we need iteration and prototypes to try out and evaluate
  - But iteration can get trapped in local maxima, designs that have no simple improvements, but are not good
  - Theory and models can help give good start points

### Usability

- It is a **measure of the effectiveness, efficiency and satisfaction** with which specified *user can achieve specified goals in a particular environment*.

**It asks the following:**

- Is effective to use
- Is efficient to use
- Is safe to use
- Has good utility
- Is easy to learn
- Is easy to remember how to use

### Issues in Design

- Who are the users?
- What do we mean by needs?
- How do generate alternative design?
- How do we choose between designs?

**User Centered Development**

- Data Collection
- Data Analysis
- User Modeling
- Design
- Prototyping
- Evaluation

- In **User Modeling**, a computational model for how people perform tasks and solve problems based on **psychological principles.**

- In **design phase**, all the analysis are used to design a system in interface design, it shows how this product present itself. And in **interaction design**, it tells how this product should work.

- **Rapid prototypes** are early and inexpensive ways to identify usability problems before committing lots of resources. In **interaction design** main concern here is with usability. So, *using prototypes we can better understands the needs of user*.

- In **usability testing and evaluation phase**, user perform a variety of task with a prototype (or other system) while observes record notes on what each user does and speaks.

- *Typical tests are conducted with one user at a time or two users working together.* **Testing** may include collecting data on the paths users take to do tasks, the errors they make, when and where they are confused or frustrated, how fast they do a task, whether they *succeed in doing the task, and how satisfied they are with the experience.*

- The **goal of most usability testing** is to uncover any problems that users may encounter so those problems can be fixed.

**Interaction Styles**

Interaction can be seen as **dialogue between the computer and the user**. Some applications have distinct styles of interaction.

We can identify some **common interaction styles :**

- Command Line Interface
- Menus
- Question/answer and query dialogue
- Form- fills and spreadsheet
- WIMP(Windows, Icon, Menus, Pointer)

**Windows**

**Areas of the screen** that behave as if they were independent terminals

- can **contain text or graphics**
- can be **moved or resized**
- can **overlap and obscure each other**, or can be laid out next to one another
- **scrollbars** allow the user to move the contents of the window up and down or from side to side
- **title bars** describe the name of the window

**Icons**

Small picture or image, used to represent some object in the interface, often a window. **Windows** *can be closed to this small representation (iconized) allowing many windows to be accessible*. **Icons can be many and various – highly stylized or realistic representations.**

**Pointers**

**Important component**, since WIMP style relies on pointing things such as icons and menu items.

- usually achieved with mouse

- joystick, trackball, cursor keys or keyboard shortcuts are also used.

- wide variety

**LESSON 2: DESIGN EVALUATION**

**Design Evaluation**
is a rich theme that includes **ideas, methods and resources.**

**What is Evaluation in HCI**?
- The role is to *access designs and test systems to ensure that they do what we expect and satisfy user* requirement.

- **Evaluation** is a process that reviews a programme critically.

**Goal of Evaluation**

1. Evaluate the quality and usability of the features of the program.
2. Determine the contact experience for the users.
3. Find any possible problems with the system.

**Forms of Design Evaluation**

1. **Expert Analysis** – The assessment process followed requires a designer and/or specialists in the evaluation.

   It attempts to consider any factor leading to the use of difficulties or to contradict proven cognitive principles.

2. **User Participation** – this includes a variety of analytical techniques, experimental techniques, quantitative techniques, physiological research survey techniques.

**Approaches to Expert Analysis**

1. **Cognitive Walkthrough** - is a usability evaluation method used to assess how easy a system or product is for new users to understand and operate. The process **involves stepping through the interface and tasks from the perspective of a user who is unfamiliar with the system**. The main goal is to identify potential usability issues and determine if the interface supports user goals and understanding.

**Scenario**: Cognitive Walkthrough for an E-Commerce Website
**Task:** Purchase a product.
**Steps**:

Search for a product.
Click "Add to Cart."
Go to the cart and click "Checkout."
Enter shipping and payment information.
Click "Place Order."

2. **Heuristic Evaluation** – Heuristic is a rule that can be used to guide a decision on design or criticize a decision that has already been made. Heuristic Evaluation is a usability inspection method used to identify usability problems in a user interface (UI). It **involves a small group of evaluators examining the interface and assessing it against a set of established usability principles**, **known as heuristics.**

**Example Scenario**: Heuristic Evaluation for a Shopping App
**Task**: Purchase a product.

Findings:

Visibility of system status:
Problem: No loading indicator during checkout.
Severity: 3 (Major problem).
Recommendation: Add a progress bar.
Consistency and standards:
Problem: Some buttons use "Buy" while others use "Purchase."
Severity: 2 (Minor problem).
Recommendation: Use consistent terminology.

Error prevention:
Problem: No confirmation step before placing an order.
Severity: 4 (Catastrophe).
Recommendation: Add a review step before finalizing the order.

Common Severity Rating Scale (0–4)

Here's a typical scale used to rate the severity of usability problems:

| Rating | Severity Level | Description | Action Recommended |
|---|---|---|---|
| 0 | Not a Problem | No usability issue detected. | No action needed. |
| 1 | Cosmetic Problem | Minor issue that doesn't impact task completion but might affect aesthetics. | Fix if time allows. |
| 2 | Minor Usability Problem | Causes slight inconvenience but doesn't prevent task completion. | Fix if resources permit. |
| 3 | Major Usability Problem | Significantly hinders task completion or frustrates users. | Fix as a high priority. |
| 4 | Usability Catastrophe (Critical Issue) | Prevents task completion, causing severe frustration or abandonment. | Fix immediately. |

2. **Model-based Evaluation** - is a **usability assessment method** that utilizes computational or theoretical models to analyze and predict user behavior and performance within a system or interface.

   - This approach is particularly useful for understanding how users might interact with a design, identifying potential issues, and making informed design decisions based on modeled user interactions.

4. **Evaluation Dependent on Analysis** - refers to a method of assessing the usability and effectiveness of a system or product based on data-driven insights gathered from various analyses. This approach emphasizes the use of empirical data and analytical techniques to inform the evaluation process, rather than relying solely on subjective judgments or intuitive assessments.

**Evaluation Dependent on Analysis**

**a. Experimental Assessment**

- Control review of various aspects of the digital behavior.
- Needs hypothesis testing.
- results of behavioral approaches are linked to specific factors.

b. Experimental Design
c. Analysis
d. Team Architecture
e. Intergroup design

**Test Types**

a. **Parametric**
   - Assumes normal distribution
   - Robust
   - Potent

b. **Non Parametric**
   - Do not assume natural
   - Less efficient
   - More stable distribution

c. **Table of Contingency**
   - Information recognition by different attributes
   - Number of pieces of information in each category

**Laboratory Research vs. Field Research**

**A. Laboratory Research** – it refers that is **completed within the laboratory**. Researchers or scientists do some experiments and run tests inside the laboratory, which is **a controlled environment.**

B. **Field Research** – usually **brought out when researchers are researching on the topics associated with people**. Researchers could use forms like questionnaires and surveys to get fresh data from the natural environment.

**Types of User-Based Evaluation**

**1. Observational Methods** – methods are often applied in the psychological, scientific, political and economic sectors.

a. **Think Aloud** -is a technique used to gain insight into someone's thought process by having them verbalize their thoughts, decisions, and feelings as they perform a task. This approach is **commonly used in user experience (UX) research, cognitive psychology, and**

**problem-solving studies** to understand how people interact with products, solve problems, or make decisions.

Example:
If someone is testing a mobile app, they might say things like:

"I'm looking for the settings icon… I guess it might be in the top right corner."
"Hmm, this button doesn't do what I expected. Maybe I clicked the wrong thing?

**b**. **Cooperative Evaluation** is a user-centered evaluation technique used in user experience (UX) and human-computer interaction (HCI) studies. It involves **collaborating directly with users to assess and improve a product's usability** by encouraging open communication and feedback during testing sessions.

Example:
Imagine testing a new e-commerce website:
A user tries to find and purchase a product. As they navigate, they might say:
"I'm not sure where to find the filter options. Shouldn't they be on the left side?"
The evaluator can respond:
"What kind of filter options would you expect to see there?"
This dialogue helps uncover usability issues and user expectations quickly.

**C. Protocol Analysis**
- Is a research method used to study people's thought processes by analyzing their verbalized thoughts as they perform a task. It involves **collecting and examining the "think-aloud protocols"** *—the step-by-step explanations provided by individuals as they solve problems, make decisions, or interact with a product.*

This method is widely used in cognitive psychology, user experience (UX) research, education, and problem-solving studies to gain insights into how people think, reason, and make decisions.

**Example:** A protocol analysis session for testing a new website might go like this:

User: "Okay, I want to find the support page… I think it should be in the footer. Oh, it's not there, maybe I'll try the menu instead."
Researcher: Records and later analyzes this protocol to understand where the navigation could be improved.

**D. Automated Analysis**
- refers to the use of software tools and algorithms to process, interpret, and extract meaningful insights from data without manual intervention. This approach is **applied across various fields, including data science, user experience (UX), cybersecurity, finance**, and more, to handle large datasets quickly and efficiently.

**Types of Automated Analysis**

**1.Text Analysis:** Uses **NLP** to extract insights from documents, social media, or customer feedback.
Example: Sentiment analysis to gauge customer satisfaction.
**2.Data Analysis:** Applies **statistical methods** and machine learning to find patterns, trends, or anomalies.
Example: Predicting sales trends based on historical data.
**3.Security Analysis**: Uses **algorithms** to detect threats, unusual behavior, or vulnerabilities.
Example: Intrusion detection systems in cybersecurity.
**4.UX and Behavior Analysis:** Analyzes **user interactions** with websites or apps to identify usability issues.
Example: Heatmaps showing where users click most often.

**E.Post-task Walkthroughs**

- **Post-task walkthrough** is a usability testing technique in which participants review and reflect on their actions immediately after completing a task. This method **helps researchers gain a deeper understanding of users' thought processes, challenges, and feelings about the task.**

**Example Scenario**:
Imagine testing an e-commerce website:
**Task**: A user tries to find and purchase a product.
**Walkthrough**: After completing the task, the researcher shows a recording of the session and asks:
"You hesitated here—what was going through your mind?"
User: "I wasn't sure if this button meant 'add to cart' or 'buy now'."
Insight: Suggests the need for clearer button labels.

**Types of User-Based Evaluation**

2. **Query Techniques -** refer to various methods and strategies used to formulate and execute queries for retrieving information from databases, search engines, or other information systems. These techniques **help users specify what data they need, refine search results, and extract meaningful insights effectively.**

Query techniques are commonly used in database management, information retrieval, web search, and natural language processing (NLP) to **improve the accuracy and relevance of the results.**

**Interviews** - the interviewer asks some questions to the user based on produced questionnaires. It is also actual, personal and relatively low-cost.

**Questionnaires** - are a popular method for gathering quantitative and qualitative feedback from users about their experiences, preferences, and perceptions when interacting with systems, interfaces, or products.

**C. Physiological and Direct Recording -** are methods used in Human-Computer Interaction (HCI) and usability studies to capture objective data about users' physical and emotional states as they interact with systems, interfaces, or products. These techniques **help researchers understand how users truly feel and react, beyond what they verbally express.**

**D. Experimental Methods** - refer to systematic approaches for testing hypotheses by manipulating variables and observing their effects on participants. This method aims to **establish cause-and-effect relationships by *controlling external factors and comparing different conditions.***

**LESSON 3: EXPERIMENTAL EVALUATION AND EMPIRICAL METHODS**

**Hypothesis Testing** – a technique used in statistics in order to make decision based on the gathered information from an experimental data. It is created by **Ronald Fisher, Jerzy Neyman, Karl Pearson and Egon Pearson.**

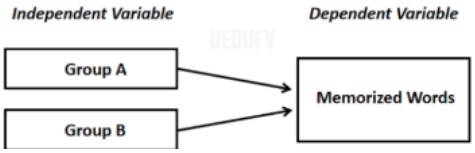
**Variables** is a symbol representing information that can take on different values or change.

1. An **independent variable** is the type of variable which can be changed by the researcher/experimenter and is assumed to obtain direct effect on the dependent variable.
2. A **dependent variable** is a type of subjected variable on a test which is measured in the experiment, and dependent on independent variable.

Example:

Let's assume you test the difference between two groups (Group A and Group B) of respondents and how well they can remember ten given words after one day.

- What is being tested is the number of words being memorized, which is the study's dependent variable in this case.




## Levels of Measurement in a Dependent Variable

1. **Nominal Measurement** - categorizes data without any order or ranking.

   **Characteristics:**

   - Categories are **mutually exclusive** (each item fits into one group only).
   - No numerical or logical order.

   **Examples:**

   - Gender: Male, Female, Non-binary
   - Eye color: Blue, Brown, Green
   - Types of fruit: Apple, Banana, Mango

2. **Ordinal Measurement** - categorizes data **with** a meaningful order, but **without equal intervals** between categories.

   **Characteristics:**

   - Can rank the data (e.g., 1st, 2nd, 3rd).
   - Distance between ranks is **not measurable**.

   **Examples:**

   - Customer satisfaction: Very satisfied, Satisfied, Neutral, Dissatisfied, Very dissatisfied
   - Education level: High school, Bachelor's, Master's, PhD
   - Pain level: Mild, Moderate, Severe

3. **Metric Measurement** ***(Also called Interval or Ratio, collectively referred to as "scale" or "continuous")*** - Data is measured on a scale with **equal intervals**; allows for meaningful arithmetic operations.

   - **Two types:**
     - **Interval scale**: Has equal intervals, **but no true zero**.
       - Example: *Temperature in Celsius or Fahrenheit (0°C doesn't mean "no temperature").*
     - **Ratio scale**: Has equal intervals **and a true zero**.
       - Example: *Height, weight, age, income*

**Data Analysis** is a data collection, transformation, and refurbishment and modelling method with the intention of unveiling the required information.

## Phases of Data Analyzation

1. **Data Cleaning** – is the analysis of data is the documents are cross-checked duplication eliminated, and column separation is performed to refurbish the obtained raw data in each different source.
2. **Quality analysis** - is the process of evaluating and assessing the quality of a product, service, or process to ensure it meets predefined standards, requirements, or customer expectations. It plays a key role in quality management and continuous improvement.
3. **Quality of measurement** refers to how accurate, reliable, and valid a measurement is. In simple terms, it's

about how *good* your measurement is — how well it captures what you're trying to measure.

4. **Analysis** - is the process of breaking something down into smaller parts to understand it better, find patterns, or draw conclusions. It's basically a deep dive into information, data, systems, or situations to make sense of them.

**Choosing Participants and Sample Size**

**Participants** are the people (or sometimes groups or entities) who take part in a study, experiment, survey, program, or activity.

**Random assignment** is a method used in experiments to randomly place participants into different groups, such as a treatment group and a control group.

**Sample size** is the number of participants, items, or observations included in a study, survey, or experiment.

The sample size is at least 100. Most statisticians conclude that the required sample size is 100 to obtain any significant result.

**Three Types of Experimental Designs**

1. **Individual Test / In Groups** (also known as **Independent Groups Design**)

**Individual Test**: Participants are assigned to one group or condition and tested individually. The results from each participant are treated separately.

**In Groups (Independent Groups Design):**

- Participants are randomly assigned to one of two or more groups (e.g., control group or experimental group).
- Each group experiences a different condition of the experiment.
- Participants in one group are not exposed to the conditions of the other groups.

2. **Repeated Measures Design**

- Participants are exposed to every condition of the experiment.
- Every participant acts as their own control group, experiencing all levels of the independent variable.
- Comparison is made within the same participant across different conditions.

3. **Matched Pairs Design**

- Participants are paired based on similarities in characteristics (e.g., age, IQ, baseline performance) before the experiment.
- One participants from each pair is randomly assigned to each group or condition (e.g., experimental or control group).
- Ensures that the groups are as similar as possible at the start of the experiment.